

HR & Payroll Analytics Dashboard

Project Report – Microsoft Power BI

An interactive Business Intelligence project for analyzing employee, salary, payroll, department and workforce information.

1. Project Overview

The **HR & Payroll Analytics Dashboard** is a Microsoft Power BI project developed to transform employee and payroll data into interactive and meaningful business insights. The report combines employee information, salary and payroll measures, department analysis, gender distribution, designation, experience, budget and time-based trends in a single reporting solution.

The project contains four analytical pages: **HR & Payroll Overview**, **Payroll Analysis**, **Employee Analysis**, and **Department Analysis**. Interactive slicers allow users to filter the report by department, location, gender, designation and year, while DAX measures provide dynamic KPI calculations.

2. Project Objectives

- Analyze total employee strength and workforce composition.
- Understand employee distribution across departments and designations.
- Compare male and female employees.
- Calculate average salary and average employee experience.
- Analyze total payroll, bonuses, deductions and salary-related measures.
- Compare payroll and salary across departments and locations.
- Analyze department budgets and average department size.
- Study employee and payroll trends across years and months.
- Provide an interactive dashboard for HR and management decision-making.

3. Dataset Information

The Power BI model uses employee, salary/payroll, department and calendar information. The main salary/payroll table is **fact_salary_dataset**, supported by employee and department tables and a calendar dimension for time-based analysis.

Table	Purpose / Important Fields
fact_salary_dataset	EmployeeID, Salary, Bonus, Budget, HRA, Net Salary and payroll-related fields
employee_dataset	Employee information including Gender and Designation
dept_dataset	Department information including DeptName
Dim calander	Date, Month, Month Name, Quarter and Year for time analysis

4. Data Cleaning & Preparation

- Imported the required employee, salary, department and calendar data into Power BI.
- Reviewed table structures, column names and data types before building visuals.
- Used Power Query for data transformation and preparation.
- Ensured employee identifiers could be used for distinct employee counting.
- Prepared date, month and year fields for time-based analysis.
- Created or used a numeric Month field to sort Month Name chronologically.
- Created DAX measures for employee count, salary, budget and other KPIs.
- Formatted salary, payroll and budget values for clear dashboard presentation.
- Established relationships between the relevant tables for interactive filtering.

5. Tools & Technologies Used

- **Microsoft Power BI:** dashboard development, visualization and interactive reporting.
- **Power Query:** data cleaning, transformation and preparation.
- **DAX:** dynamic measures and KPI calculations.
- **Data Modeling:** relationships between employee, salary, department and calendar tables.
- **Power BI Visuals:** cards, bar charts, column charts, line charts, donut charts, tables and slicers.

6. Dashboard Features

- **KPI Cards:** display important metrics such as employee count, payroll, bonus, deductions, average salary, budget and department count.
- **Interactive Slicers:** allow filtering by department, location, gender, designation, manager and year depending on the dashboard page.
- **Department Analysis:** compares employee count, salary and budget-related metrics between departments.
- **Payroll Analysis:** analyzes payroll by department, month and location and displays bonus information.
- **Employee Analysis:** provides gender, designation, joining-year and department analysis.
- **Drill-through:** supports detailed employee-level analysis from summary views.
- **Trend Analysis:** line charts show changes across months or years.
- **Dynamic Reporting:** KPI cards and visuals respond to selected slicers.

7. Important Calculations

Total Employee Count: DISTINCTCOUNT of EmployeeID.

Total Departments: DISTINCTCOUNT of DeptName.

Average Salary: AVERAGE of the salary field.

Average Experience: AVERAGE of YearsOfService where available.

Average Department Size: average of distinct employee counts across departments.

Total Budget: sum of department budget values, with care taken to avoid double counting when budgets repeat.

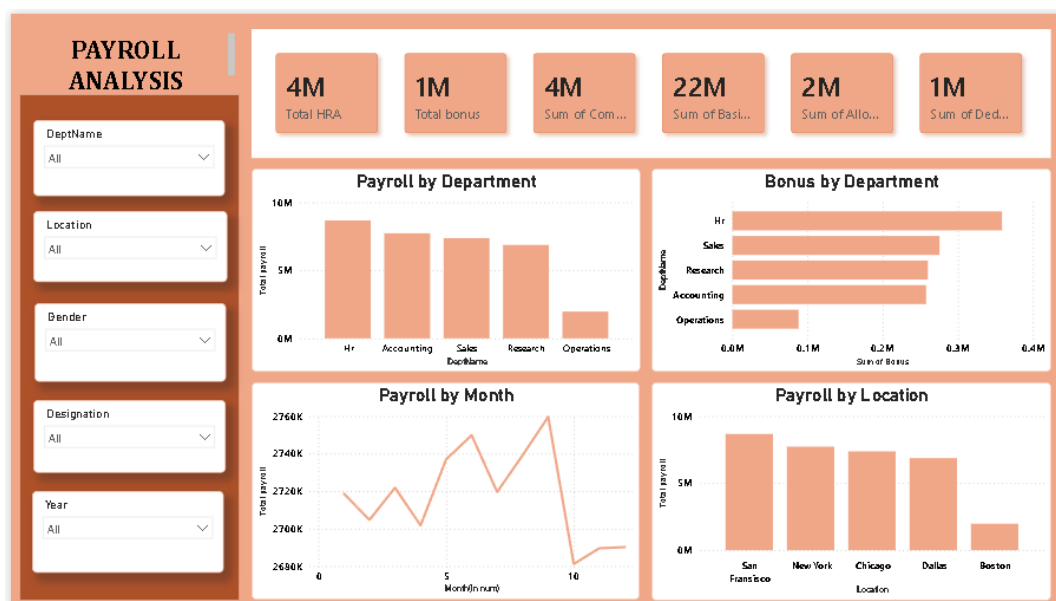
8. Dashboard Screenshots & Analysis

Dashboard 1 – HR & Payroll Overview



This page demonstrates the interactive Power BI visual layout, KPI cards, filters and charts used for the selected analysis.

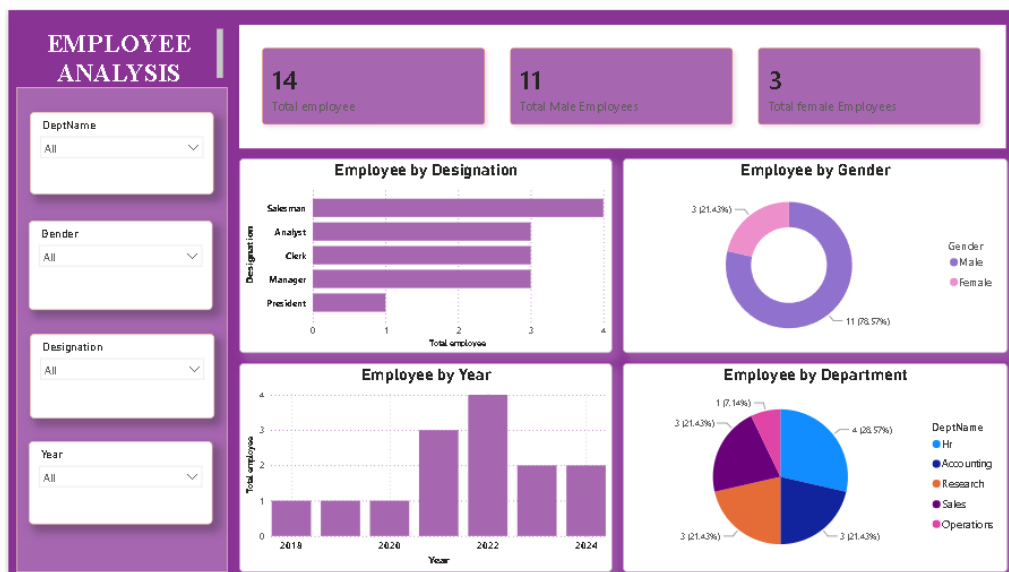
Dashboard 2 – Payroll Analysis



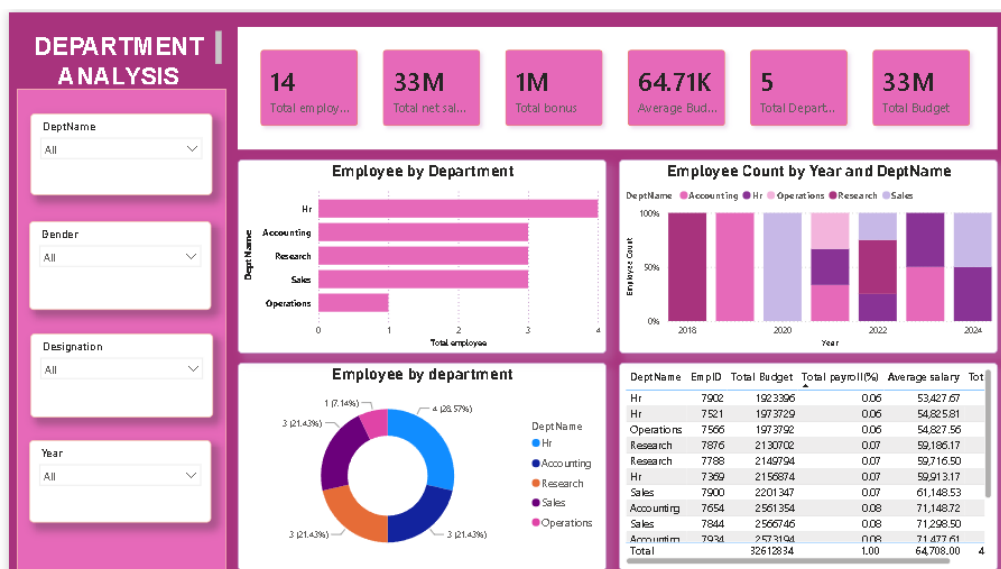
This page demonstrates the interactive Power BI visual layout, KPI cards, filters and charts used for the selected analysis.

9. Dashboard Screenshots & Key Insights

Dashboard 3 – Employee Analysis



Dashboard 4 – Department Analysis



10. Key Insights

- The dashboard provides a consolidated view of employee and payroll information across multiple analytical dimensions.
- Department and designation visuals make it possible to identify how the workforce is distributed across roles and teams.
- The gender analysis shows the composition of the employee population and can be used to support workforce diversity analysis.
- Payroll visuals allow comparison of payroll across departments, locations and months.
- Average salary and department-level salary comparisons help identify differences in compensation across organizational units.
- Department analysis combines employee count, budget and salary measures to support resource planning.
- Year and month analysis enables users to identify changes and patterns over time.
- Interactive slicers make it possible to perform focused analysis for a selected department, gender, designation or year.

11. Conclusion

The **HR & Payroll Analytics Dashboard** successfully demonstrates how Power BI can be used to convert structured HR and payroll data into an interactive Business Intelligence solution. The four dashboard pages provide complementary views of the organization: an overall HR and payroll summary, detailed payroll analysis, employee analysis and department analysis.

The project demonstrates practical skills in **Power BI, Power Query, DAX, data modeling, data visualization and dashboard design**. The final report can support HR teams and management in understanding workforce composition, compensation, payroll distribution and departmental trends.

Project: HR & Payroll Analytics Dashboard **Technology:** Microsoft Power BI **Author:** Anjali Patnaik